Project Proposal: M-CORE

Team Name: elgoog | **GitHub Repository:** https://github.com/Jeswanthkavi/

1. Problem Statement

Modern cities are getting hit harder by two intertwined climate threats: extreme urban heatwaves and rapid surface flash flooding. Right now, local governments and emergency responders evaluate these crises using completely disconnected tools.

The core issue is that urban heat and urban flooding share the exact same physical triggers. High concrete density, lack of green canopy, and non-porous asphalt trap massive amounts of solar radiation (creating dangerous heat islands) while simultaneously preventing rainwater from soaking into the ground (causing flash floods).

While academia acknowledges this "compound risk," existing tools fail at an operational level:

- **Current weather models operate at coarse scales (10km+),** making them practically useless for ward- or street-level decisions.
- **Traditional hydrodynamic simulations are impractically slow,** requiring expensive proprietary software, detailed drainage pipe blueprints, and intensive computing power.
- **Sensor networks are costly,** leaving massive coverage gaps across vulnerable neighborhoods.

There is an urgent need for an agile, software-driven platform that translates open earth-observation data into immediate, block-by-block compound risk forecasts.

2. Who It Helps

- **Municipal Government & City Planners:** Helps city managers allocate emergency resources (e.g., cooling centers, mobile water pumps, barricades) precisely where they are needed most, rather than issuing blanket city-wide warnings.
- **Urban First Responders:** Gives emergency teams a 24-to-72-hour head start to deploy assets before a heatwave or severe storm strikes.
- **Vulnerable Micro-Communities:** Protects dense, under-resourced urban neighborhoods that suffer disproportionately from lack of green space and inadequate stormwater infrastructure.

3. Proposed Solution

We built **M-CORE** (Micro-Urban Compound Heat-Flood Risk Engine), a software-only predictive platform that breaks cities down into high-resolution micro-grids (10m–100m). Instead of relying on expensive hardware or physical sensors, M-CORE pulls freely available satellite imagery, digital elevation models, street geometries, and dynamic weather forecasts.

It evaluates local heat stress, surface flood susceptibility, and their non-linear compounding impact in real time. Through transparent, explainable AI factors, M-CORE tells city officials not just *which* street block is at critical risk over the next 24–72 hours, but *why*—allowing targeted, high-impact intervention.

4. High-Level Architecture

The diagram below outlines the detailed flow of data through M-CORE, from multi-source spatial ingestion through feature engineering, dual-model prediction, compound interaction modeling, down to explainable action outputs:

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|                                STAGE 1: DATA INGESTION                            |
|  +-------------------+  +-------------------+  +---------------+  +-----------------+
|  | Sentinel-2 Bands |  | Copernicus DEM  |  | OpenStreetMap |  | Open-Meteo API  |
|  | (10m Imagery)    |  | (Elevation Map) |  | (Geometries)  |  | (Forecast Rain/T)|
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|                             STAGE 2: FEATURE ENGINEERING                          |
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|  |   Pipeline 1: Computer Vision        |  |  Pipeline 2: Terrain & Weather     |  |
|  |   - SegFormer / U-Net Segmentation   |  |  - Elevation Depressions & Slope   |  |
|  |   - Impervious %, Canopy %, NDVI     |  |  - Runoff Coeff & Rain/Temp Vectors|  |
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|                            STAGE 3: MULTI-HAZARD ML ENGINES                       |
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|  | Heat Risk Model (XGBoost/LightGBM)|  | Flood Susceptibility Model (XGB/RF)  |  |
|  | Maps land cover & forecast temp   |  | Maps runoff, slope & rain intensity  |  |
|  | Output: Heat Score (0 - 100)      |  | Output: Flood Score (0 - 100)        |  |
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|                          STAGE 4: COMPOUND ML ENGINE & XAI                        |
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|  | Non-Linear Risk Fusion Engine (Heat Score + Flood Score Interaction)       |  |
|  | Evaluates non-linear compounding risk spike (e.g., Heat: 85 + Flood: 88 = 94) |  |
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|  | SHAP Explainability Engine (XAI)                                           |  |
|  | Factor Attribution: (+28% Impervious, +22% Low Elevation, +24% Heavy Rain)   |  |
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|                             STAGE 5: MUNICIPAL ACTION OUTPUT                      |
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|  | Interactive GIS Dashboard (Streamlit + Leaflet.js)                         |  |
|  | Block-level alerts & actionable directives (e.g., Pre-position pumps)      |  |
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- **Stage 1: Data Ingestion** — Pulls raw, open-access spatial and dynamic data streams for city micro-grids.
- **Stage 2: Feature Engineering Pipelines** — Computer Vision models segment land-use parameters, while geospatial scripts compute slope, runoff, and forecast vectors.
- **Stage 3: Multi-Hazard ML Prediction** — Parallel ML algorithms generate isolated score metrics for Heat and Flood risk.
- **Stage 4: Compound ML & XAI** — Synthesizes dual hazards into a unified compounding score and runs SHAP attributions to explain risk drivers.
- **Stage 5: Municipal Output** — Renders precise warnings and tactical operational directives onto an interactive map dashboard.

5. Tech Stack

Layer	Technology / Tools	Function
Computer Vision	PyTorch, OpenCV, SegFormer, U-Net	Land-cover segmentation & surface feature extraction from satellite rasters
Machine Learning & XAI	XGBoost, LightGBM, Scikit-learn, SHAP	Heat, flood, and compound risk modeling with factor explanations
Geospatial Processing	GeoPandas, Rasterio, GDAL, PostGIS	Micro-grid discretization, spatial alignment, coordinate re-projection
Data Sources	Sentinel-2, Copernicus DEM, OpenStreetMap, Open-Meteo API	Open-access satellite, elevation, infrastructure, and weather forecast streams
Backend & Frontend	FastAPI, Streamlit, Folium / Leaflet.js	High-speed spatial querying, REST API services, interactive GIS map UI

6. Project Repository

GitHub Link: https://github.com/Jeswanthkavi/